

M4a: a conditional continuum LPPL theorem for removal of the spatial cutoff in $P(\varphi)_2$

Abstract

This note proves the continuum Cauchy step of the spatial-cutoff programme in an explicitly stated operator-localization framework and under two analytic hypotheses: a uniform Hamiltonian gap and a uniform local second moment of the interaction density. Exact finite propagation for the *actual cutoff dynamics* and the required ground-state operator domains are proved from the standard cutoff construction. Affine-path differentiability and endpoint continuity are proved from the Feynman–Kac–Nelson formula and the assumed gap. No bounded quasi-adiabatic generator is used. Instead, differentiating the state gives a connected correlation with one unbounded argument. An exact spectral filter and a gap-plus-locality estimate reduce this correlation to a summable one-dimensional cell expansion.

The conclusion is norm convergence of the full cutoff net on every local algebra, with a rate faster than every inverse power of the distance to the cutoff, local normality, translation invariance, $\mathbb{Z}_2$ -invariance for even P , and the algebraic ground-state property proved in M0. The theorem does *not* prove the two remaining hypotheses, uniqueness of all ground states, the sharp rate, or uniformity in a wavelet resolution. Those points are separated from the conditional theorem rather than hidden inside its proof.

1 Framework and hypotheses

For every bounded open $U \subset \mathbb{R}$, let $\mathfrak{A}(U)$ be the von Neumann algebra generated by the time-zero Weyl operators whose test functions have support in U . Let $O \subset \mathbb{R}$ be a bounded open interval and $O_r := \{x \in \mathbb{R} : \text{dist}(x, O) < r\}$. We use the time-zero local net $U \mapsto \mathfrak{A}(U) \subset B(\mathcal{H}_0)$ and its norm-closed quasi-local algebra $\mathfrak{A} = \overline{\bigcup_O \mathfrak{A}(O)}^{\|\cdot\|}$. The net is isotonic and local:

$$O_1 \cap O_2 = \emptyset \implies [\mathfrak{A}(O_1), \mathfrak{A}(O_2)] = 0.$$

For real $h \in C_c^\infty(\mathbb{R})$, write

$$V(h) := \int_{\mathbb{R}} h(x) : P(\varphi_0(x)) : dx.$$

The operator-localization framework used below is the following. For every real $h \in C_c^\infty(\mathbb{R})$, $V(h)$ is a specified self-adjoint operator; it is affiliated with $\mathfrak{A}(U)$ for every bounded open U containing $\text{supp } h$. If $h = \sum_{j=1}^N h_j$ and $\xi \in D(V(h)) \cap \bigcap_{j=1}^N D(V(h_j))$, then

$$V(h)\xi = \sum_{j=1}^N V(h_j)\xi.$$

These assertions are part of the standard operator-localization framework.

Let

$$\mathcal{G} := \{g \in C_c^\infty(\mathbb{R}) : 0 \leq g \leq 1, \ g = 1 \text{ on a nonempty bounded interval}\},$$

ordered by $g \preceq g'$ if $\{g = 1\}^\circ \subseteq \{g' = 1\}^\circ$. Put

$$K(g) := H_0 + V(g), \quad E(g) := \inf \text{spec } K(g), \quad H(g) := K(g) - E(g).$$

For each $g \in \mathcal{G}$, assume that 0 is a simple eigenvalue of $H(g)$ with normalized eigenvector Ω_g . Set

$$\omega_g(A) := \langle \Omega_g, A \Omega_g \rangle, \quad \beta_t^g(A) := e^{itH(g)} A e^{-itH(g)}.$$

Inner products are linear in the second argument.

This order is directed. Indeed, for $g_1, g_2 \in \mathcal{G}$, choose a bounded interval I containing $\{g_1 = 1\} \cup \{g_2 = 1\}$ and a function $g_3 \in C_c^\infty(\mathbb{R}; [0, 1])$ equal to 1 on a neighbourhood of $\bar{I}$; then $g_1, g_2 \preceq g_3$. Similarly, for every bounded interval O and $d < \infty$, one may choose $g_0 \in \mathcal{G}$ equal to 1 on O_d , and every $g \succ g_0$ is also equal to 1 on O_d .

The following are the exact hypotheses used below.

Hypothesis 1.1 (uniform gap, G). *There is $\gamma > 0$ such that, for every $g \in \mathcal{G}$,*

$$\text{spec } H(g) \subseteq \{0\} \cup [\gamma, \infty).$$

Hypothesis 1.2 (uniform local interaction moment, M). *There is $M < \infty$ such that, for every $g \in \mathcal{G}$ and every real $h \in C_c^\infty(\mathbb{R})$ with $\|h\|_\infty \leq 1$ whose support is contained in an interval of length at most 2,*

$$\|V(h)\Omega_g\| \leq M.$$

Proposition 1.3 (domain smoothing and cutoff propagation). *The standard spatial-cutoff $P(\varphi)_2$ construction has the following two consequences.*

1. *For every $g \in \mathcal{G}$ and every real $h \in C_c^\infty(\mathbb{R})$, $\Omega_g \in D(V(h))$.*
2. *For every $g \in \mathcal{G}$, every bounded interval O , and every $t \in \mathbb{R}$,*

$$\beta_t^g(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|}).$$

The propagation-speed bound is one and is independent of g .

Proof. For the first assertion, the Glimm–Jaffe semigroup-smoothing theorem [1, Ch. 7, Thm. 7.3 and Lem. 7.4] gives, for $t > 0$ and every compactly smeared Wick polynomial $V(h)$,

$$\text{Ran } e^{-tK(g)} \subseteq D(V(h)).$$

Since $K(g)\Omega_g = E(g)\Omega_g$, $\Omega_g = e^{tE(g)}e^{-tK(g)}\Omega_g \in D(V(h))$.

For propagation, the interaction automorphism $\beta_t^{I,g} = \text{Ad } e^{itV(g)}$ has speed zero. Indeed, for an observable generated by Weyl operators supported in a compact $B \Subset O$, choose $\chi \in C_c^\infty(O)$ equal to one near B . The strongly commuting decomposition

$$V(g) = V(\chi g) + V((1 - \chi)g)$$

has $e^{itV(\chi g)} \in \mathfrak{A}(O)$ and $e^{itV((1-\chi)g)} \in \mathfrak{A}(B)'$, whence $\beta_t^{I,g}(A) \in \mathfrak{A}(O)$. Strong density extends this to every $A \in \mathfrak{A}(O)$. The free dynamics $\beta_t^0 = \text{Ad } e^{itH_0}$ has speed one by the finite-propagation property of the Klein–Gordon equation. Therefore the n th Trotter approximant

$$(\beta_{t/n}^0 \circ \beta_{t/n}^{I,g})^n(A)$$

belongs to $\mathfrak{A}(O_{|t|})$. The Trotter product formula gives strong convergence of these approximants to $\beta_t^g(A)$, and the von Neumann algebra $\mathfrak{A}(O_{|t|})$ is strongly closed. This is the standard “free speed one + interaction speed zero” proof [1, Ch. 4, Prop. 4.1.1 and Thm. 4.1.2]. $\square$

Proposition 1.4 (affine-path regularity from FKN). *Let $g, g' \in \mathcal{G}$ be equal to one on a common nonempty interval, set $g_s = (1-s)g + sg'$, $v = g' - g$, and abbreviate*

$$K_s = K(g_s), \quad E_s = E(g_s), \quad H_s = K_s - E_s, \quad P_s = |\Omega_s\rangle\langle\Omega_s|, \quad Q_s = 1 - P_s.$$

Under Hypothesis 1.1, $s \mapsto P_s$ is norm-continuous on $[0, 1]$ and C^1 on $(0, 1)$. Locally on $(0, 1)$, normalized ground-state vectors can be chosen C^1 in Hilbert norm and in the parallel gauge. They then obey

$$\dot{E}_s = \langle \Omega_s, V(v) \Omega_s \rangle, \quad \dot{\Omega}_s = -H_s^{-1} Q_s V(v) \Omega_s. \quad (\text{FKN})$$

In particular, $s \mapsto \omega_{g_s}(A) = \text{Tr}(P_s A)$ is continuous on $[0, 1]$ for every $A \in B(\mathcal{H}_0)$.

Proof. Let $m_0 > 0$ be the free mass and fix $T > 0$. On free Euclidean path space, with time-slice maps J_0, J_T , define

$$\mathcal{U}_s = \int_0^T \int_{\mathbb{R}} g_s(x) : P(\varphi(t, x)) : dx dt, \quad w_s = e^{-\mathcal{U}_s}, \quad \mathcal{U}_v = \mathcal{U}_1 - \mathcal{U}_0.$$

For bounded time-zero functions f, h , the Feynman–Kac–Nelson formula is

$$\langle f, e^{-TK_s} h \rangle = \int \overline{J_T f} (J_0 h) w_s d\mu_0. \quad (1.1)$$

Put

$$q_T = \frac{2}{1 - e^{-m_0 T}}, \quad r_T = \frac{q_T}{q_T - 1}.$$

Here is the full derivation of the time-slice product estimate. Let ν_0 be the time-zero Gaussian marginal of μ_0 , and let $P_T = e^{-TH_0}$ be its Markov semigroup. Since

$$r_T = \frac{2}{1 + e^{-m_0 T}}, \quad p_T := \frac{2}{r_T} = 1 + e^{-m_0 T}, \quad p'_T = 1 + e^{m_0 T},$$

p'_T is conjugate to p_T and $(p'_T - 1)/(p_T - 1) = e^{2m_0 T}$. Nelson hypercontractivity gives $\|P_T F\|_{p'_T} \leq \|F\|_{p_T}$. Hence the Markov property and Hölder's inequality give

$$\begin{aligned} \|(J_T f)(J_0 h)\|_{r_T}^{r_T} &= \int |h|^{r_T} P_T(|f|^{r_T}) d\nu_0 \\ &\leq \| |h|^{r_T} \|_{p_T} \|P_T(|f|^{r_T})\|_{p'_T} \\ &\leq \| |h|^{r_T} \|_{p_T} \| |f|^{r_T} \|_{p_T} = \|h\|_2^{r_T} \|f\|_2^{r_T}. \end{aligned}$$

Taking the r_T th root yields

$$\|(J_T f)(J_0 h)\|_{r_T} \leq \|f\|_2 \|h\|_2. \quad (1.2)$$

The underlying hypercontractivity and transfer-matrix facts are [5, Thms. V.10 and V.12].

The spacetime-smeared Wick polynomial $\mathcal{U}_v$ belongs to $L^p(\mu_0)$ for every finite p . Since $g, g' \geq 0$ and P is semibounded, the exponential estimate [5, Thm. V.7] gives

$$e^{-a\mathcal{U}_0}, e^{-a\mathcal{U}_1} \in L^1(\mu_0) \quad (a < \infty).$$

Because $\mathcal{U}_s = (1-s)\mathcal{U}_0 + s\mathcal{U}_1$, Hölder's inequality gives

$$\|w_s\|_a^a \leq \left(\int e^{-a\mathcal{U}_0} d\mu_0 \right)^{1-s} \left(\int e^{-a\mathcal{U}_1} d\mu_0 \right)^s. \quad (1.3)$$

Thus $\sup_{0 \leq s \leq 1} \|w_s\|_a < \infty$ for each finite a .

If $s_n \rightarrow s$, then $\mathcal{U}_{s_n} \rightarrow \mathcal{U}_s$ in every finite L^p , so $w_{s_n} \rightarrow w_s$ in probability. Applying (1.3) with an exponent strictly larger than a makes $(|w_{s_n}|^a)_n$ uniformly integrable. Vitali's theorem therefore yields

$$\|w_{s_n} - w_s\|_a \rightarrow 0 \quad (1.4)$$

for every finite a . Fix $q = q_T$ and choose finite p, a with $q^{-1} = p^{-1} + a^{-1}$. Whenever $s, s+h \in [0, 1]$, the scalar fundamental theorem of calculus gives, pointwise in path space,

$$\frac{w_{s+h} - w_s}{h} = -\mathcal{U}_v \int_0^1 w_{s+\theta h} d\theta.$$

Consequently, for $0 < s < 1$,

$$\left\| \frac{w_{s+h} - w_s}{h} + \mathcal{U}_v w_s \right\|_q \leq \|\mathcal{U}_v\|_p \sup_{0 \leq \theta \leq 1} \|w_{s+\theta h} - w_s\|_a \rightarrow 0.$$

Thus $s \mapsto w_s$ is continuous from $[0, 1]$ to L^{q_T} and is C^1 from $(0, 1)$ to L^{q_T} , with derivative $-\mathcal{U}_v w_s$. The derivative is continuous because

$$\|\mathcal{U}_v(w_s - w_r)\|_{q_T} \leq \|\mathcal{U}_v\|_p \|w_s - w_r\|_a \rightarrow 0.$$

Equations (1.1)–(1.2), first for bounded f, h and then by L^2 density, imply

$$\|e^{-TK_s} - e^{-TK_r}\| \leq \|w_s - w_r\|_{q_T},$$

and the identical estimate for difference quotients proves that $s \mapsto L_s := e^{-TK_s}$ is norm-continuous on $[0, 1]$ and norm- C^1 on $(0, 1)$.

The largest spectral value of L_s is the simple eigenvalue $\lambda_s = e^{-TE_s}$. Hypothesis 1.1 gives

$$\text{spec } L_s \setminus \{\lambda_s\} \subset [0, e^{-T\gamma\lambda_s}].$$

Since $\lambda_s = \|L_s\|$, it is continuous; hence the displayed relative separation is locally bounded below by a positive absolute distance. The Riesz integral around λ_s now proves that P_s is norm-continuous on $[0, 1]$ and C^1 on $(0, 1)$. Near any $s_0 \in (0, 1)$,

$$\tilde{\Omega}_s = \frac{P_s \Omega_{s_0}}{\|P_s \Omega_{s_0}\|}$$

is C^1 after shrinking the neighbourhood. A C^1 phase change puts it in the parallel gauge $\langle \Omega_s, \dot{\Omega}_s \rangle = 0$. The identity $\lambda_s = \langle \Omega_s, L_s \Omega_s \rangle$ then shows explicitly that λ_s , and hence $E_s = -T^{-1} \log \lambda_s$, is C^1 on $(0, 1)$.

It remains to identify the derivative. The cutoff Hamiltonians have the common operator core $\mathcal{C} = C^\infty(H_0)$ and, on that core,

$$K_s \xi = K_{s_0} \xi + (s - s_0) V(v) \xi, \quad \xi \in \mathcal{C},$$

by [5, Thm. V.12(a)]. For fixed $\xi \in \mathcal{C}$, differentiate $\langle K_s \xi, \Omega_s \rangle = E_s \langle \xi, \Omega_s \rangle$. Here $E_s = -T^{-1} \log \lambda_s$ is C^1 , and Proposition 1.3 gives $\Omega_s \in D(V(v))$. Self-adjointness of $V(v)$ therefore yields

$$\langle H_s \xi, \dot{\Omega}_s \rangle = \langle \xi, (\dot{E}_s - V(v)) \Omega_s \rangle \quad (\xi \in \mathcal{C}).$$

Since $\mathcal{C}$ is a core for H_s , the definition of the adjoint implies $\dot{\Omega}_s \in D(H_s)$ and

$$H_s \dot{\Omega}_s = (\dot{E}_s - V(v)) \Omega_s.$$

Taking the inner product with Ω_s gives the first identity in (FKN); the parallel gauge and $\|H_s^{-1}Q_s\| \leq \gamma^{-1}$ give the second. Finally,

$$|\operatorname{Tr}((P_s - P_r)A)| \leq \|P_s - P_r\|_1 \|A\| \leq 2\|P_s - P_r\| \|A\|,$$

because $P_s - P_r$ has rank at most two. This proves the endpoint assertion. $\square$

For the final ground-state identification and invariances, we also use the already established facts

$$\alpha_t(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|}), \quad \alpha_t(A) = \beta_t^g(A) \quad \text{if } g = 1 \text{ on } O_{|t|}, \quad (\text{P})$$

$$\omega_{\tau_a g} = \omega_g \circ \tau_{-a}, \quad (\text{T})$$

where $(\tau_a g)(x) = g(x - a)$ and the same symbol denotes the spatial automorphism of $\mathfrak{A}$.

2 An exact spectral filter

The proof needs an L^1 real-time representation of H^{-1} on the gapped subspace. The following construction gives more decay than is needed for convergence.

Lemma 2.1 (exact filter). *For every $\gamma > 0$ there is $W_\gamma \in L^1(\mathbb{R})$ such that, with*

$$\widehat{W}_\gamma(E) := \int_{\mathbb{R}} W_\gamma(t) e^{itE} dt,$$

one has

$$\widehat{W}_\gamma(0) = 0, \quad \widehat{W}_\gamma(E) = \frac{1}{E} \quad \text{for } |E| \geq \gamma.$$

Moreover, for every integer $q \geq 0$,

$$\int_{\mathbb{R}} (1 + |t|)^q |W_\gamma(t)| dt < \infty.$$

Consequently, if $H \geq 0$ and $\operatorname{spec} H \subseteq \{0\} \cup [\gamma, \infty)$ with $Q = 1 - P_{\ker H}$, then

$$H^{-1}Q = \int_{\mathbb{R}} W_\gamma(t) e^{itH} Q dt$$

as a norm-convergent Bochner integral.

Proof. Choose an even $\chi \in C^\infty(\mathbb{R}; [0, 1])$ with $\chi(E) = 0$ for $|E| \leq \gamma/2$ and $\chi(E) = 1$ for $|E| \geq \gamma$. Define

$$\widehat{W}_\gamma(E) = \begin{cases} \chi(E)/E, & E \neq 0, \\ 0, & E = 0. \end{cases}$$

This is a smooth odd function. For every $k \geq 1$, its k th derivative is integrable: it is smooth on a compact transition region and is a constant multiple of E^{-k-1} outside that region.

Define the inverse Fourier transform first as an oscillatory integral,

$$W_\gamma(t) := \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itE} \widehat{W}_\gamma(E) dE.$$

For $t \neq 0$, oddness gives

$$W_\gamma(t) = -\frac{i}{\pi} \int_0^\infty \sin(tE) \frac{\chi(E)}{E} dE.$$

This is uniformly bounded for $0 < |t| \leq 1$: the integral over $[0, \gamma]$ is bounded absolutely, and after the substitution $u = |t|E$ the integral over $[\gamma, \infty)$ is a tail of $\int_0^\infty \sin u/u \, du$, whose partial integrals are uniformly bounded by Dirichlet's test. For $|t| \geq 1$, integration by parts k times in the oscillatory integral gives

$$|W_\gamma(t)| \leq \frac{\|\widehat{W}_\gamma^{(k)}\|_{L^1}}{2\pi|t|^k} \quad (k \geq 1),$$

because every boundary term vanishes. Taking $k > q + 1$ proves the moment assertion and in particular $W_\gamma \in L^1$. Fourier inversion in tempered distributions, followed by continuity of the Fourier transform of an L^1 function, gives $\mathcal{F}W_\gamma = \widehat{W}_\gamma$ pointwise.

The spectral theorem now yields

$$\int_{\mathbb{R}} W_\gamma(t) e^{itH} Q \, dt = \widehat{W}_\gamma(H) Q = H^{-1} Q.$$

The integral converges in operator norm because $W_\gamma \in L^1$ and $\|e^{itH} Q\| \leq 1$. $\square$

Put

$$T_\gamma(r) := \int_{|t|>r} |W_\gamma(t)| \, dt.$$

Lemma 2.1 implies, for every $q \geq 0$,

$$T_\gamma(r) \leq C_{q,\gamma} (1+r)^{-q}. \quad (1)$$

3 Clustering with one unbounded local observable

We first isolate the scalar spectral-separation estimate used in M3.

Lemma 3.1 (spectral separation). *Let*

$$F(t) = \int_{[\gamma, \infty)} e^{itE} \, d\rho(E), \quad G(t) = \int_{(-\infty, -\gamma]} e^{itE} \, d\sigma(E),$$

where ρ and σ are complex measures with $|\rho|(\mathbb{R}), |\sigma|(\mathbb{R}) \leq M_0$. If $F(t) = G(t)$ for $|t| < R$ and $\gamma R \geq 1$, then

$$|F(0)| \leq 2M_0 e^{-\gamma R/2}.$$

Proof. For $a, \varepsilon > 0$, set

$$f_{a,\varepsilon}(t) := \frac{1}{2\pi i} \frac{e^{-t^2/(2a)}}{t - i\varepsilon}.$$

Residue calculus followed by convolution with the Fourier transform of the Gaussian gives

$$\check{f}_{a,\varepsilon}(E) := \int_{\mathbb{R}} f_{a,\varepsilon}(t) e^{itE} \, dt = \int_0^\infty \sqrt{\frac{a}{2\pi}} e^{-a(E-E')^2/2} e^{-\varepsilon E'} \, dE'.$$

Hence $0 \leq \check{f}_{a,\varepsilon} \leq 1$ and $\check{f}_{a,\varepsilon}(E) \rightarrow \Phi(\sqrt{a}E)$ as $\varepsilon \downarrow 0$, where Φ is the standard normal distribution function. Also

$$\int_{|t| \geq R} |f_{a,\varepsilon}(t)| \, dt \leq \frac{a}{\pi R^2} e^{-R^2/(2a)}.$$

Indeed, $|t - i\varepsilon| \geq |t|$ and $\int_R^\infty e^{-t^2/(2a)} \, dt \leq (a/R) e^{-R^2/(2a)}.$

Using Fubini, the equality $F = G$ on $(-R, R)$, and the total-variation bounds, then letting $\varepsilon \downarrow 0$, gives

$$\left| \int \Phi(\sqrt{a}E) d\rho(E) - \int \Phi(\sqrt{a}E) d\sigma(E) \right| \leq \frac{2M_0a}{\pi R^2} e^{-R^2/(2a)}.$$

For $x \geq 0$, $1 - \Phi(x) \leq \frac{1}{2}e^{-x^2/2}$ and $\Phi(-x) \leq \frac{1}{2}e^{-x^2/2}$. The support conditions therefore imply

$$|F(0)| \leq M_0 e^{-a\gamma^2/2} + \frac{2M_0a}{\pi R^2} e^{-R^2/(2a)}.$$

Taking $a = R/\gamma$ and using $\gamma R \geq 1$ gives

$$|F(0)| \leq M_0 e^{-\gamma R/2} \left(1 + \frac{2}{\pi \gamma R} \right) \leq 2M_0 e^{-\gamma R/2}.$$

□

Lemma 3.2 (one unbounded argument). *Let $H \geq 0$ have normalized simple ground state Ω and $\text{spec } H \subseteq \{0\} \cup [\gamma, \infty)$. Suppose its dynamics $\beta_t = \text{Ad } e^{itH}$ satisfies exact finite propagation. Let $B \subset \mathbb{R}$ be compact, let $X = X^*$ be affiliated with $\mathfrak{A}(U)$ for every bounded open neighbourhood U of B , let $\Omega \in D(X)$, and let $A \in \mathfrak{A}(O)$, where O is a bounded open interval. If $R := \text{dist}(B, O) > 0$ and $\gamma R \geq 1$, then*

$$|\langle X\Omega, A\Omega \rangle - \langle X\Omega, \Omega \rangle \langle \Omega, A\Omega \rangle| \leq 2\|X\Omega\| \|A\| e^{-\gamma R/2}. \quad (2)$$

Proof. Let $P_0 = |\Omega\rangle\langle\Omega|$ and define

$$F(t) := \langle X\Omega, (e^{itH} - P_0)A\Omega \rangle, \quad G(t) := \langle A^*\Omega, (e^{-itH} - P_0)X\Omega \rangle.$$

The spectral theorem represents F and G by complex measures supported in $[\gamma, \infty)$ and $(-\infty, -\gamma]$, respectively, with total variations at most $M_0 := \|X\Omega\| \|A\|$.

Fix $|t| < R$ and choose a bounded open neighbourhood U of B disjoint from $O_{|t|}$. Finite propagation gives $\beta_t(A) \in \mathfrak{A}(O_{|t|}) \subseteq \mathfrak{A}(U)'$. Affiliation of X with $\mathfrak{A}(U)$ implies that every element of $\mathfrak{A}(U)'$ preserves $D(X)$ and commutes with X there. Since $e^{-itH}\Omega = \Omega$,

$$\langle X\Omega, e^{itH}A\Omega \rangle = \langle \Omega, X\beta_t(A)\Omega \rangle = \langle \Omega, \beta_t(A)X\Omega \rangle = \langle A^*\Omega, e^{-itH}X\Omega \rangle.$$

The two P_0 terms are also equal: both are $\langle X\Omega, \Omega \rangle \langle \Omega, A\Omega \rangle$, because $\langle X\Omega, \Omega \rangle \in \mathbb{R}$. Hence $F(t) = G(t)$ for $|t| < R$. Lemma 3.1 applies, and $F(0)$ is the expression on the left of (2). □

4 Differentiating the ground-state expectation

Lemma 4.1 (state derivative). *Assume Hypothesis 1.1. Let $g, g' \in \mathcal{G}$ be equal to one on a common nonempty interval, put $g_s = (1-s)g + sg'$, and, for $0 < s < 1$, write*

$$V := V(g' - g), \quad E_s := E(g_s), \quad H_s := H(g_s), \quad P_s := |\Omega_s\rangle\langle\Omega_s|, \quad Q_s := 1 - P_s.$$

Then

$$\dot{E}_s = \langle \Omega_s, V\Omega_s \rangle, \quad (3)$$

$$\dot{\Omega}_s = -H_s^{-1}Q_sV\Omega_s, \quad (4)$$

and, for every self-adjoint $A \in B(\mathcal{H}_0)$,

$$\frac{d}{ds}\omega_{g_s}(A) = -2 \operatorname{Re} \int_{\mathbb{R}} W_\gamma(t) \left[\langle V\Omega_s, \beta_t^{g_s}(A)\Omega_s \rangle - \langle V\Omega_s, \Omega_s \rangle \omega_{g_s}(A) \right] dt. \quad (5)$$

The integral is absolutely convergent.

Proof. The first two identities are exactly Proposition 1.4; in particular all products with the unbounded operator V are defined by Proposition 1.3.

For self-adjoint A ,

$$\frac{d}{ds}\omega_{g_s}(A) = 2 \operatorname{Re} \langle \dot{\Omega}_s, A\Omega_s \rangle = -2 \operatorname{Re} \langle V\Omega_s, H_s^{-1}Q_s A\Omega_s \rangle.$$

Insert Lemma 2.1. Since $e^{itH_s}A\Omega_s = \beta_t^{g_s}(A)\Omega_s$ and $P_s A\Omega_s = \omega_{g_s}(A)\Omega_s$, this is exactly (5). Absolute convergence follows from

$$|\langle V\Omega_s, e^{itH_s}Q_s A\Omega_s \rangle| \leq \|V\Omega_s\| \|A\|$$

and $W_\gamma \in L^1$. □

5 The cell estimate and the Cauchy theorem

Choose $\rho \in C_c^\infty((-1, 1))$ with $0 \leq \rho \leq 1$ and $\rho > 0$ on $[-\frac{1}{2}, \frac{1}{2}]$, put

$$D(x) := \sum_{k \in \mathbb{Z}} \rho(x - k), \quad \eta_j(x) := \frac{\rho(x - j)}{D(x)}.$$

The sum defining D is locally finite, $D(x) > 0$, and D is smooth and one-periodic. Hence $(\eta_j)_{j \in \mathbb{Z}}$ is a smooth partition of unity satisfying

$$0 \leq \eta_j \leq 1, \quad \operatorname{supp} \eta_j \subset (j - 1, j + 1), \quad \sum_{j \in \mathbb{Z}} \eta_j = 1.$$

For $v = g' - g$, set $h_j = \eta_j v$ and $V_j = V(h_j)$. Only finitely many h_j are nonzero, $V(v) = \sum_j V_j$, and Hypothesis 1.2 gives

$$\|V_j \Omega_s\| \leq M \quad (0 \leq s \leq 1). \quad (6)$$

Proposition 5.1 (one path). *Assume Hypotheses 1.1 and 1.2. Let O be a bounded interval and suppose $g = g' = 1$ on O_d , where $d \geq d_0 := 4 + 2/\gamma$. Then*

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq 32M \sum_{n \geq \lfloor d \rfloor} \left(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \right), \quad (7)$$

Proof. It is enough to take self-adjoint $A \in \mathfrak{A}(O)$ with $\|A\| \leq 1$, because $\omega_g - \omega_{g'}$ is hermitian. Since $v = 0$ on O_d , every nonzero h_j is supported at distance at least d from O . Put $B_j = \operatorname{supp} h_j$ and $r_j = \operatorname{dist}(B_j, O)$.

For $0 < s < 1$, the inequalities $0 \leq g, g' \leq 1$ imply

$$g_s(x) = 1 \iff g(x) = g'(x) = 1.$$

Thus $g_s \in \mathcal{G}$ and $g_s = 1$ on O_d for all $s \in [0, 1]$, so both hypotheses apply along the entire path.

For fixed s, j , define the vector correlation

$$C_{j,s}(t) := \langle V_j \Omega_s, \beta_t^{g_s}(A) \Omega_s \rangle - \langle V_j \Omega_s, \Omega_s \rangle \omega_{g_s}(A).$$

If $|t| \leq r_j/2$, Proposition 1.3 gives $\beta_t^{g_s}(A) \in \mathfrak{A}(O_{|t|})$, whose distance from B_j is at least $r_j/2$. Moreover, $\gamma r_j/2 \geq 1$ because $r_j \geq d \geq 2/\gamma$. Lemma 3.2, applied to $X = V_j$ and $\beta_t^{g_s}(A)$, together with (6), yields

$$|C_{j,s}(t)| \leq 2M e^{-\gamma r_j/4}. \quad (8)$$

For arbitrary t , Cauchy–Schwarz gives the uniform bound

$$|C_{j,s}(t)| \leq |\langle V_j \Omega_s, \beta_t^{g_s}(A) \Omega_s \rangle| + |\langle V_j \Omega_s, \Omega_s \rangle| |\omega_{g_s}(A)| \leq 2M. \quad (9)$$

Substituting $V = \sum_j V_j$ into (5), splitting the t -integral at $|t| = r_j/2$, and using (8)–(9), gives

$$\left| \frac{d}{ds} \omega_{g_s}(A) \right| \leq 4M \sum_j \left(\|W_\gamma\|_1 e^{-\gamma r_j/4} + T_\gamma(r_j/2) \right).$$

Write $O = (a, b)$. Because $d \geq 4$ and $B_j \subset [j-1, j+1]$, a nonempty B_j cannot have points on both sides of O_d . If B_j is on the right and $r_j \in [n, n+1)$, compactness gives $x_j \in B_j$ with $x_j - b = r_j$. Since $|j - x_j| \leq 1$, necessarily $j \in [b+n-1, b+n+2]$, an interval of length 3 containing at most four integers. The same count holds on the left. Consequently at most eight indices occur in each shell $r_j \in [n, n+1)$. Both functions $r \mapsto e^{-\gamma r/4}$ and $r \mapsto T_\gamma(r/2)$ are decreasing. Since $r_j \geq d$, the last display is therefore at most

$$32M \sum_{n \geq \lfloor d \rfloor} \left(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \right).$$

For $0 < \varepsilon < 1/2$, integrate the last derivative bound over $[\varepsilon, 1-\varepsilon]$. Proposition 1.4 gives norm continuity of P_s at $s = 0, 1$, hence continuity of $\omega_{g_s}(A) = \text{Tr}(P_s A)$ there. Letting $\varepsilon \downarrow 0$ and taking the supremum over the self-adjoint unit ball proves the proposition. $\square$

Define

$$\Psi_\gamma(d) := 32M \sum_{n \geq \lfloor d \rfloor} \left(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \right) \quad (d \geq d_0), \quad (10)$$

and set $\Psi_\gamma(d) := \max\{2, \Psi_\gamma(d_0)\}$ for $0 \leq d < d_0$. This is a bounded decreasing function; the value 2 gives the trivial bound for the norm-distance of two states when Proposition 5.1 is unavailable.

Lemma 5.2 (decay of the comparison function). *$\Psi_\gamma(d) \rightarrow 0$ as $d \rightarrow \infty$. More precisely, for every integer $N \geq 1$ there is $C_{N,\gamma,M} < \infty$ such that*

$$\Psi_\gamma(d) \leq C_{N,\gamma,M} (1+d)^{-N}.$$

Proof. The exponential series in (10) has an exponentially decaying tail. For the second series, Lemma 2.1 and Markov's inequality give, for every $q \geq 1$,

$$T_\gamma(r) \leq r^{-q} \int_{\mathbb{R}} |t|^q |W_\gamma(t)| dt.$$

Choose $q = N + 2$ and sum n^{-q} over $n \geq \lfloor d \rfloor$. $\square$

Theorem 5.3 (conditional continuum LPPL). *Assume Hypotheses 1.1 and 1.2. Then there is a state ω_∞ on $\mathfrak{A}$ such that, for every bounded interval O ,*

$$\lim_{g \in \mathcal{G}} \|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| = 0.$$

For every g with $g = 1$ on O_d ,

$$\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \Psi_\gamma(d), \quad (11)$$

and Ψ_γ obeys Lemma 5.2. The restriction $\omega_\infty|_{\mathfrak{A}(O)}$ is normal for every O .

If (P) holds, ω_∞ is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense of M0. If (T) holds, ω_∞ is translation invariant. If P is even and every ω_g is $\mathbb{Z}_2$ -invariant, then so is ω_∞ .

Proof. Fix O and $\varepsilon > 0$. By Lemma 5.2, choose d with $\Psi_\gamma(d) < \varepsilon$. Choose $g_0 \in \mathcal{G}$ with $g_0 = 1$ on O_d . If $g, g' \succ g_0$, then $g = g' = 1$ on O_d , so Proposition 5.1 gives

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| < \varepsilon.$$

Thus $(\omega_g|_{\mathfrak{A}(O)})_{g \in \mathcal{G}}$ is Cauchy in the Banach space $\mathfrak{A}(O)^*$. Its limit is a state ω_O . The limits are compatible under isotony, so they define a positive norm-one functional on $\bigcup_O \mathfrak{A}(O)$, which extends uniquely by norm continuity to a state ω_∞ on $\mathfrak{A}$.

To prove (11), first suppose $d \geq d_0$. Fix $g = 1$ on O_d and let g' tend to infinity through the cofinal tail of cutoffs which are also 1 on O_d . Proposition 5.1 applies to every such g' , and norm convergence permits passage to the limit. If $0 \leq d < d_0$, the assertion follows from $\|\omega_g - \omega_\infty\| \leq 2 \leq \Psi_\gamma(d)$.

Each $\omega_g|_{\mathfrak{A}(O)}$ is normal. Normal functionals form a norm-closed subspace of $\mathfrak{A}(O)^*$; explicitly, if $0 \leq A_i \uparrow A$ in $\mathfrak{A}(O)$ and $\|\omega_g - \omega_\infty\|_{\mathfrak{A}(O)^*} < \delta$, then

$$0 \leq \omega_\infty(A) - \omega_\infty(A_i) \leq 2\delta\|A\| + \omega_g(A - A_i).$$

First let i increase, using normality of ω_g , and then let $\delta \downarrow 0$. Hence $\omega_\infty|_{\mathfrak{A}(O)}$ is normal.

Choose an increasing sequence $(\tilde{g}_n)_{n \geq 1} \subset \mathcal{G}$ such that $\tilde{g}_n = 1$ on $(-n, n)$. For every bounded interval I ,

$$d(\tilde{g}_n, I) := \text{dist}(I, \mathbb{R} \setminus \{\tilde{g}_n = 1\}^\circ) \longrightarrow \infty, \quad \|(\omega_{\tilde{g}_n} - \omega_\infty)|_{\mathfrak{A}(I)}\| \longrightarrow 0.$$

These are exactly the two hypotheses of M0, Theorem 4.2. Its exact spectral-passage argument, using (P), therefore implies that ω_∞ is an α -ground state.

For translation invariance, fix $a \in \mathbb{R}$ and $A \in \mathfrak{A}(O)$. Translation is an order automorphism of $\mathcal{G}$, so both g and $\tau_a g$ tend cofinally to infinity. By (T),

$$\omega_\infty(A) = \lim_g \omega_{\tau_a g}(A) = \lim_g \omega_g(\tau_{-a} A) = \omega_\infty(\tau_{-a} A).$$

Density extends this from local A to $\mathfrak{A}$. The $\mathbb{Z}_2$ assertion follows directly by taking the local norm limit of invariant states. $\square$

Remark 5.4 (what the theorem does not prove). Theorem 5.3 proves uniqueness of the limit of the specified cutoff family. It does not show that every locally normal ground state of $(\mathfrak{A}, \alpha)$ equals ω_∞ . Identification with a separately constructed Euclidean vacuum also requires either a direct comparison theorem or an independent uniqueness result. The super-polynomial function Ψ_γ is an explicit valid rate furnished by the chosen exact filter; no claim of a sharp e^{-2md} rate is made.

6 Remaining obligations

Proposition 1.3 closes exact cutoff propagation and every vector-domain assertion. Proposition 1.4 closes affine-path differentiability and endpoint continuity under the already assumed gap. Theorem 5.3 therefore reduces the remaining continuum Cauchy step to two statements:

1. prove the full-net uniform physical gap of Hypothesis 1.1 in a precisely specified noncritical parameter regime;
2. prove the uniform local second moment in Hypothesis 1.2 by a valid ultraviolet estimate.

The discharge audit and explicit reductions are in `remaining.tex`. In particular, the covariance-weighted Poincaré induction previously proposed for the moment bound is invalid and is not used. A wavelet resolution requires additional comparison maps, a uniform interacting Lieb–Robinson estimate, and ultraviolet convergence, and is therefore a separate theorem rather than part of Theorem 5.3.

References

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